Superoptimal continued fractions

Slade Sanderson

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Generalised continued fractions

Definition

A **generalised continued fraction** (GCF) is an expression

$$[d_0; e_0/d_1, e_1/d_2, \dots] = d_0 + \cfrac{e_0}{d_1 + \cfrac{e_1}{d_2 + \ddots}}, \qquad e_n, d_n \in \mathbb{Q}, \quad e_n \neq 0.$$

Its **convergents** are $P_n/Q_n := [d_0; e_0/d_1, \dots, e_{n-1}/d_n]$.
If $\lim_n P_n/Q_n = x \in \mathbb{R}$, call this a GCF-**expansion** of x.

(Semi-) Regular continued fractions

Certain conditions guarantee convergence of $[d_0; e_0/d_1, e_1/d_2, \dots]$:

Example

Regular continued fractions (RCFs) are GCFs satisfying

(i) $e_n = 1$,

(ii) $d_n \in \mathbb{Z}$ with $d_n > 0$ for $n > 0$.

For RCFs, use special notation $x = [d_0; d_1, d_2, \dots] := [d_0; 1/d_1, 1/d_2, \dots]$, $p_n/q_n := P_n/Q_n$.

(Semi-) Regular continued fractions

Certain conditions guarantee convergence of $[d_0; e_0/d_1, e_1/d_2, \dots]$:

Example

Semi-regular continued fractions (SRCFs) are GCFs satisfying

(i) $e_n = \pm 1$,

(ii) $d_n \in \mathbb{Z}$ with $d_n > 0$ for $n > 0$,

(iii) $e_n + d_n \geq 1$, strict infinitely often.

For RCFs, use special notation $x = [d_0; d_1, d_2, \dots] := [d_0; 1/d_1, 1/d_2, \dots]$, $p_n/q_n := P_n/Q_n$.

Optimal continued fractions

Definition

The **approximation coefficient** of $x \notin \mathbb{Q}$ and $p/q \in \mathbb{Q}$ (reduced) is $\Theta(x, p/q) := q^2|x - p/q|$.

Fact

For a.e. x, any SRCF*-exp'n. of x satisfies both*

$$\textit{(i)}\ \sup_k \Theta(x, P_k/Q_k) \geq 1/2 \qquad \& \qquad \textit{(ii)}\ \limsup_k n(k)/k \leq \log 2/\log \varphi,$$

where $q_{n(k)} \leq Q_k < q_{n(k)+1}$ and $\varphi = \frac{\sqrt{5}+1}{2}$.

Optimal continued fractions (OCF; Bosma, 1987) are SRCFs satisfying

$$\text{(i)}\ \Theta(x, P_k/Q_k) \leq 1/2 \quad \forall k \qquad \& \qquad \text{(ii)}\ \lim_k n(k)/k = \log 2/\log \varphi \quad \text{a.s.}$$

Superoptimal continued fractions

The OCFs are optimal among SRCFs, but not necessarily among GCFs.

Definition

Let $\varepsilon, C > 0$. Call a GCF-exp'n of x (ε, C)-**superoptimal** if its convergents satisfy both

$$\text{(i) } \Theta(x, P_k/Q_k) \le \varepsilon \quad \forall k \qquad \& \qquad \text{(ii) } \liminf_k n(k)/k \ge C,$$

where $q_{n(k)} \le Q_k < q_{n(k)+1}$.

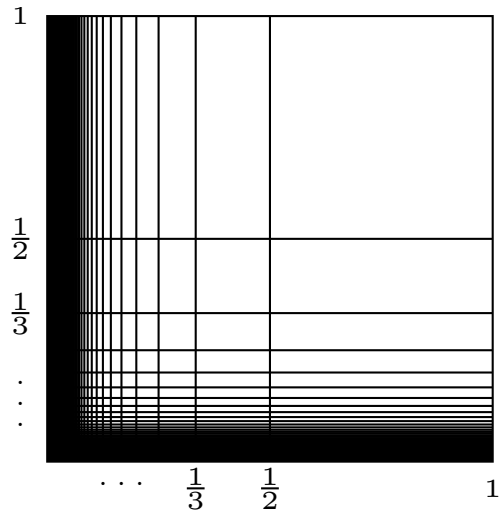
The natural extension of the Gauss map

Define $\mathcal{G} : [0,1]^2 \to [0,1]^2$, where for $z = (x, y)$, $x \neq 0$,

$$\mathcal{G}(z) := \left(M^{-1} \cdot x, M^T \cdot y\right) \quad \text{with} \quad M = M(z) = \begin{pmatrix} 0 & 1 \\ 1 & \lfloor 1/x \rfloor \end{pmatrix}$$

Symbolically,

$$\begin{aligned} &\mathcal{G}([0; a_1, a_2, \dots], [0; b_1, b_2, \dots]) \\ =&([0; a_2, a_3, \dots], [0; a_1, b_1, b_2, \dots]). \end{aligned}$$



Fact

$\mathcal{G}$ has an ergodic, a.c.i.p. μ with density $d\mu/d\lambda = 1/((\log 2)(1+xy)^2)$.

Induced transformations and Δ-GCFs

Let $\Delta \subset [0,1]^2$, $\mu(\Delta) > 0$. Ergodicity, $\mathcal{G}$-invariance of $\mu \implies$ $\mathcal{G}^n(z) \in \Delta$ i.o. a.s. Restricting to such z, define $\mathcal{G}_\Delta : [0,1]^2 \to \Delta$ by $\mathcal{G}_\Delta(z) := \mathcal{G}^{j(z)}(z)$ where

$$j(z) := \min\{n > 0 \mid \mathcal{G}^n(z) \in \Delta\}.$$

Then $\mathcal{G}_\Delta(z) = (M_\Delta^{-1} \cdot x, M_\Delta^T \cdot y)$, where

$$M_\Delta = M_\Delta(z) = \begin{pmatrix} r(z) & p(z) \\ s(z) & q(z) \end{pmatrix} := M(z) \cdots M(\mathcal{G}^{j(z)-1}(z)).$$

Remark

If also $\mu(\partial\Delta) = 0$, then for a.e. $x \in (0,1)$, $z_k := \mathcal{G}_\Delta^k(x,0)$ is defined $\forall\ k$.

The Δ-GCF of x is $[d_0^\Delta; e_0^\Delta/d_1^\Delta, e_1^\Delta/d_2^\Delta, \dots]$, where

$$\begin{pmatrix} e_0^\Delta \\ d_0^\Delta \end{pmatrix} := \begin{pmatrix} -\det(M_\Delta(z))/s(z) \\ r(z)/s(z) \end{pmatrix}, \quad \begin{pmatrix} e_{k+1}^\Delta \\ d_{k+1}^\Delta \end{pmatrix} := \begin{pmatrix} -\det(M_\Delta(z_{k+1}))s(z_k)/s(z_{k+1}) \\ q(z_k) + s(z_k)r(z_{k+1})/s(z_{k+1}) \end{pmatrix}.$$

Main result

Theorem (S. 2026+)

Define $f : [0,1]^2 \to [0,1]$ *by* $f(\alpha,\beta) = \beta/(1+\alpha\beta)$. *If both*

$$(a) \quad \Delta \subset f^{-1}([0,\varepsilon]) \quad \& \quad (b) \quad 0 < \mu(\Delta) = \mu(\mathring{\Delta}) \leq 1/C,$$

then the Δ-GCF *of* x *is a* GCF*-exp'n of* x *and is a.s.* (ε, C)*-superoptimal.*

Ideas.

(i) Contraction (Seidel, 1855): $(P_k^\Delta/Q_k^\Delta)_k = (p_{n(k)}/q_{n(k)})_k$ where $n(k)$ are those n s.t. $\mathcal{G}^{n+1}(x,0) \in \Delta$

(ii) $f(\mathcal{G}^{n+1}(x,0)) = \Theta(x, p_n/q_n)\ \forall\ n \implies \Theta(x, P_k^\Delta/Q_k^\Delta) \leq \varepsilon\ \forall\ k$

(iii) μ-equidistribution of $(\mathcal{G}^k(x,0))_k$ (Jager, 1986) $\implies$

$$\lim_k \frac{k}{n(k)} = \lim_k \frac{1}{n(k)} \sum_{j=1}^{n(k)} \mathbf{1}_\Delta(\mathcal{G}^j(x,0)) = \mu(\Delta) \leq 1/C \quad \text{a.s.} \quad \square$$

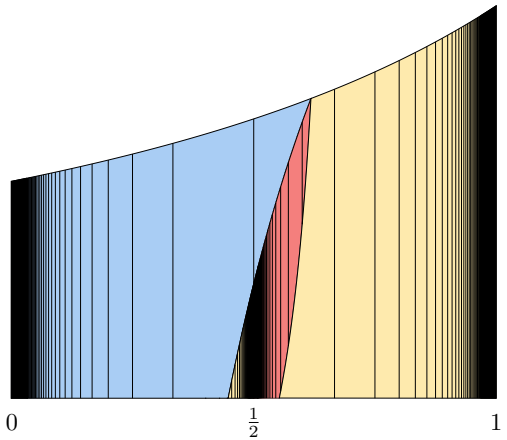
Hurwitz-Borel continued fractions

Theorem (Hurwitz, 1891; Borel 1903)

For any $x \notin \mathbb{Q}$, there are infinitely many $p/q \in \mathbb{Q}$ s.t. $\Theta(x, p/q) < 1/\sqrt{5}$. In fact, for any $n \in \mathbb{N}$, $\min\{\Theta(x, p_{n+j}/q_{n+j}) \mid j = 0, 1, 2\} < 1/\sqrt{5}$.

Setting $\Delta = f^{-1}([0, 1/\sqrt{5}))$, the Δ-GCF of x is defined *for all* $x \in (0,1) \setminus \mathbb{Q}$, is $(1/\sqrt{5}, \sqrt{5}\log 2)$-superoptimal a.s., and its convergents are *precisely* Hurwitz's p/q.

$$M_\Delta(z) = \begin{cases} \left(\begin{smallmatrix} 0 & 1 \\ 1 & a \end{smallmatrix}\right) & \text{if } x < \frac{1}{\sqrt{5}-y},\ \left\lfloor \frac{1}{x} \right\rfloor = a \\ \left(\begin{smallmatrix} 1 & a \\ 1 & a+1 \end{smallmatrix}\right) & \text{if } \frac{\sqrt{5}(y+1)-1}{(\sqrt{5}+1)y+\sqrt{5}} < x,\ \left\lfloor \frac{x}{1-x} \right\rfloor = a \\ \left(\begin{smallmatrix} 1 & a \\ 2 & 2a+1 \end{smallmatrix}\right) & \text{if } \frac{1}{\sqrt{5}-y} \le x \le \frac{1}{2},\ \left\lfloor \frac{x}{1-2x} \right\rfloor = a \\ \left(\begin{smallmatrix} 1 & a+1 \\ 2 & 2a+1 \end{smallmatrix}\right) & \text{otherwise},\ \left\lfloor \frac{1-x}{2x-1} \right\rfloor = a \end{cases}$$



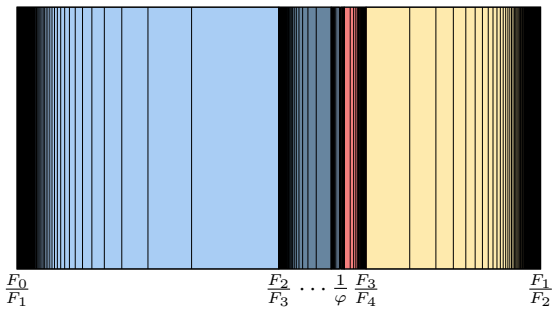
Fibonacci continued fractions

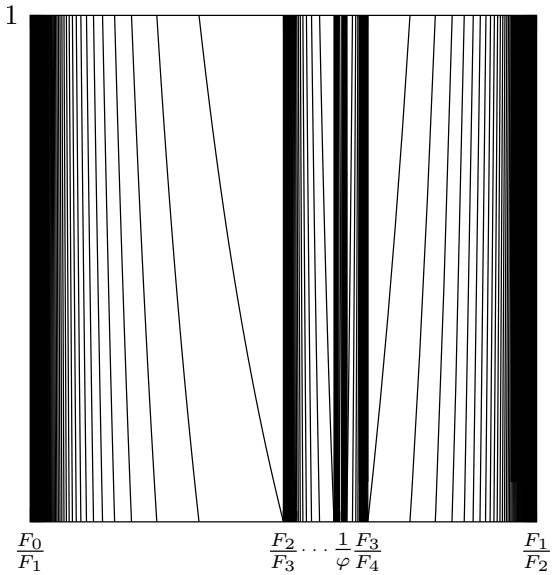
Set $\Delta = [0,1] \times [0,1/2]$. The Δ-GCF is defined $\forall\ x \notin \cup_n G^{-n}(\{0, 1/\varphi\})$, $G(x) := \{1/x\}$, and is a.s. $(1/2, \log 2/\log(3/2))$-superoptimal. One finds

$$M_\Delta(z) = \begin{pmatrix} F_n & aF_n + F_{n-1} \\ F_{n+1} & aF_{n+1} + F_n \end{pmatrix} \quad \text{for} \quad z = (x,y),\ x \in J_n(a),$$

where

$$J_n = \begin{cases} \left(\frac{F_n}{F_{n+1}}, \frac{F_{n+2}}{F_{n+3}}\right], & n \text{ even}, \\ \left[\frac{F_{n+2}}{F_{n+3}}, \frac{F_n}{F_{n+1}}\right), & n \text{ odd}, \end{cases} \qquad J_n(a) = \left\{ x \in J_n \;\middle|\; \left\lfloor \frac{F_{n-1} - F_n x}{F_{n+1} x - F_n} \right\rfloor = a \right\}.$$





$$M + \cfrac{1}{E + \cfrac{1}{R + \cfrac{1}{C + \cfrac{1}{I}}}}$$

References

É. Borel. Contribution à l'analyse arithmétique du continu. *Journal de Mathématiques Pures et Appliquées*, 9:329–375, 1903.

W. Bosma. Optimal continued fractions. *Nederl. Akad. Wetensch. Indag. Math.*, 49(4):353–379, 1987.

A. Hurwitz. Ueber die angenäherte Darstellung der Irrationalzahlen durch rationale Brüche. *Math. Ann.*, 39(2):279–284, 1891.

H. Jager. Continued fractions and ergodic theory, transcendental numbers and related topics. *RIMS Kokyuroko*, 599(1):55–59, 1986.

S. Sanderson. Superoptimal continued fractions. *Proc. Amer. Math. Soc.* (to appear).

L. Seidel. Bemerkungen über den Zusammenhang zwischen dem Bildungsgesetze eines Kettenbruches und der Art des Fortgangs seiner Näherungsbrüche. *Abh. der Kgl. Bayr. Akad. der Wiss., München, Zweite Klasse*, 7(3):559–602, 1855.